

Supplementary Materials of AnoMon

1 Mathematical Analysis

1.1 Analysis of AnoSketch

We provide the details of our mathematical analysis on AnoSketch. We first prove that AnoSketch achieves unbiased estimation for arbitrary flow group in Appendix 1.1.1. Then we prove that AnoSketch has smaller estimation variance than CocoSketch [1] in Appendix 1.1.2.

1.1.1 Proof of Unbiased Estimation

We will prove that AnoSketch provides unbiased estimation for any flow in Theorem 4.1, and thus the estimation for arbitrary flow group is also unbiased. Given an arbitrary flow f_i . Denote the real frequency of f_i as $F(f_i)$, and denote the estimated frequency of f_i as $\hat{F}(f_i)$. The estimated frequency consists of two parts: 1) the estimated frequency recorded in HotFilter (denoted as $\hat{F}_H(f_i)$), and 2) the estimated frequency recorded in ColdSampler (denoted as $\hat{F}_C(f_i)$). We have $\hat{F}(f_i) = \hat{F}_H(f_i) + \hat{F}_C(f_i)$. We suppose up to now, f_i has been inserted into ColdSampler (or evicted from HotFilter) t times with frequency $w_1, w_2, \dots, w_t$. Let $F_C(f_i) = \sum_{i=1}^t w_i$.

Lemma 4.1. $F(f_i) = \hat{F}_H(f_i) + F_C(f_i)$ holds for any flow f_i .

Proof. We prove this lemma by induction on the number of inserted packets k .

Base case: When AnoSketch is empty, the lemma holds obviously.

Induction step: Suppose the lemma holds for the first $k - 1$ insertions. Now we consider the k_{th} insertion. Suppose that the incoming packet is of flow f_k and it is mapped to a bucket in HotFilter recording flow ID f_j and positive votes p_j . We discuss the following three cases:

Case 1: $f_k = f_j$. In this case, only one flow f_j is affected by this insertion, so we just need to prove that the lemma holds for f_j , namely $F(f_j) = \hat{F}_H(f_j) + F_C(f_j)$. In this case, p_j is increased by one, namely $\hat{F}_H(f_j)$ is increased by one, and we do not insert any item to ColdSampler, so $F_C(f_j)$ is unchanged. Therefore, the increment on RHS is one, which is equal to the increment on LHS.

Case 2: $f_k \neq f_j$ and no replacement occurs in HotFilter. We need to prove that both $F(f_j) = \hat{F}_H(f_j) + F_C(f_j)$ and $F(f_k) = \hat{F}_H(f_k) + F_C(f_k)$ still hold after insertion. As there is no replacement in HotFilter, we do not modify p_j and just insert $(f_k, w = 1)$ to ColdSampler. For f_j , there is no change on either side of the equation, so the lemma still holds. For f_k , notice that $F_C(f_k)$ is increased by one, and $\hat{F}_H(f_k)$ remains zero. Therefore, the increment of RHS is one, which is equal to the increment of LHS. The lemma holds for f_k .

Case 3: $f_k \neq f_j$ and f_j is replaced by f_k in HotFilter. We need to prove that both $F(f_j) = \hat{F}_H(f_j) + F_C(f_j)$ and $F(f_k) = \hat{F}_H(f_k) + F_C(f_k)$ still hold after insertion. In this case, the hashed bucket in HotFilter is changed from $\langle f_j, p_j \rangle$ to $\langle f_k, 1 \rangle$, and we insert (f_j, p_j) to ColdSampler. We first consider the former equation. We can see that $F(f_j)$ is unchanged, $\hat{F}_H(f_j)$ is subtracted by p_j , and $F_C(f_j)$ is increased by p_j . Thus, both LHS and RHS are unchanged, so the equation still holds. For the latter equation, we can see that $F(f_k)$ and $\hat{F}_H(f_k)$ are both increased by one, and $F_C(f_k)$ is unchanged. Thus, both LHS and RHS are increased by one, so the equation still holds.

Conclusion: We have proved that both the base case and the induction step hold. Therefore, the lemma holds for any number of inserted packets k . □

Lemma 4.2. $\mathbb{E} [\hat{F}_C(f_i)] = F_C(f_i)$ for any flow f_i .

Proof. We prove this lemma by induction on the number of insertions to ColdSampler k .

Base case: when the ColdSampler is empty, the lemma holds obviously.

Induction step: Suppose the lemma holds for the first $k - 1$ insertions to ColdSampler. Now we consider the k_{th} insertion. Suppose that the incoming item is (f_k, w) and it is mapped to a bucket recording flow ID f_j and frequency $\mathcal{F}_j$. We update the mapped bucket $\mathcal{B}[h_2(f_k)]$ to $\langle f', F' \rangle$ as below.

$$\langle f', F' \rangle = \begin{cases} \langle f_k, \mathcal{F}_j + w \rangle, & \text{w.p. } \frac{w}{\mathcal{F}_j + w} \\ \langle f_j, \mathcal{F}_j + w \rangle, & \text{w.p. } \frac{\mathcal{F}_j}{\mathcal{F}_j + w} \end{cases} \quad (1)$$

After update, $\hat{F}_C(f_k)$ has a probability of $\frac{w}{\mathcal{F}_j + w}$ to change from 0 to $\mathcal{F}_j + w$, and has a probability of $\frac{\mathcal{F}_j}{\mathcal{F}_j + w}$ to be unchanged. Thus, $\mathbb{E}(\hat{F}_C(f_k))$ is increased by $\frac{w}{\mathcal{F}_j + w} \times (\mathcal{F}_j + w) = w$, which is equal to the increment of $F_C(f_k)$. On the other hand, $\hat{F}_C(f_j)$ has a probability of $\frac{w}{\mathcal{F}_j + w}$ to change from $\mathcal{F}_j$ to 0, and has a probability of $\frac{\mathcal{F}_j}{\mathcal{F}_j + w}$ to be increased by w . Thus, $\mathbb{E}(\hat{F}_C(f_j))$ is increased by $\frac{w}{\mathcal{F}_j + w} \cdot (-\mathcal{F}_j) + \frac{\mathcal{F}_j}{\mathcal{F}_j + w} \cdot w = 0$, and we note that $F_C(f_j)$ is also unchanged. In summary, the lemma still holds after the k_{th} insertion.

Conclusion: We have proved that both the base case and the induction step hold. Therefore, the lemma holds for any number of insertions k . $\square$

Theorem 4.1. For any flow f_i , its estimation frequency $\hat{F}(f_i)$ is unbiased, i.e., $\mathbb{E}[\hat{F}(f_i)] = F(f_i)$.

Proof. According to the query workflow of AnoSketch, we have $\hat{F}(f_i) = \hat{F}_H(f_i) + \hat{F}_C(f_i)$, where $\hat{F}_H(f_i)$ is a fixed value, and $\hat{F}_C(f_i)$ is a random variable. From Lemma 4.1 and Lemma 4.2, we have that

$$\begin{aligned} \mathbb{E}[\hat{F}(f_i)] &= \mathbb{E}[\hat{F}_H(f_i) + \hat{F}_C(f_i)] = \hat{F}_H(f_i) + \mathbb{E}[\hat{F}_C(f_i)] \\ &= \hat{F}_H(f_i) + F_C(f_i) = F(f_i) \end{aligned} \quad (2)$$

$\square$

1.1.2 Proof of Smaller Variance

We prove that AnoSketch has smaller variance for estimating arbitrary flow group than state-of-the-art CocoSketch [1]. In our analysis, we cast the problem of measuring flow group to the problem of subset sum estimation.

Variance for subset sum estimation: Let $F(f)$ be the real frequency of flow f , and $\hat{F}(f)$ be the estimated frequency of f . The estimation variance for subset sum estimation is defined as $Var := \sum_f (F(f) - \hat{F}(f))^2$. We consider the variance increment of inserting each item (f_i, w) to ColdSampler/CocoSketch: $\Delta Var := \sum_f \Delta (F(f) - \hat{F}(f))^2$.

Lemma 4.3. Suppose the mapped bucket is $\langle f_j, \mathcal{F}_j \rangle$, the increment of variance sum caused by inserting (f_i, w) is

$$\Delta Var = \sum_f \Delta (F(f) - \hat{F}(f))^2 = \begin{cases} 2w\mathcal{F}_j, & f_i \neq f_j \\ 0, & f_i = f_j \end{cases} \quad (3)$$

Proof. If $f_i = f_j$, we can directly update the mapped bucket to $\langle f_j, \mathcal{F}_j + w \rangle$, and the variance increment is 0.

If $f_i \neq f_j$, let $p = \frac{w}{\mathcal{F}_j + w}$, which is also the probability of replacing f_j with f_i in Eq.(3), and we have

$$\begin{aligned} \Delta Var &= p \left(\left(\frac{w}{p} - w \right)^2 + \mathcal{F}_j^2 \right) + (1 - p) \left(w^2 + \left(\frac{\mathcal{F}_j}{1 - p} - \mathcal{F}_j \right)^2 \right) \\ &= \frac{w^2}{p} - w^2 + \frac{\mathcal{F}_j^2}{1 - p} - \mathcal{F}_j^2 = 2w\mathcal{F}_j \end{aligned} \quad (4)$$

In this way, we prove that Eq.(3) always holds. $\square$

Theorem 4.2. For any flow f_i , AnoSketch has smaller estimation variance $Var[\hat{F}(f_i)]$ than CocoSketch [1].

Proof. From Lemma 4.3, we can see that the increment of variance sum caused by a single insertion is only related to w and $\mathcal{F}_j$. The smaller the frequency in the mapped bucket (namely $\mathcal{F}_j$), the smaller the variance increment will be. As AnoSketch uses HotFilter to filter large flows, the frequencies in the buckets of ColdSampler will become smaller. Thus, compared to basic CocoSketch, the variance increment of AnoSketch will also become smaller. $\square$

1.2 Analysis of CellSketch

We derive the error upper bound of CellSketch in Theorem 5.1. As multiple flows can collide into one histogram in CellSketch, which makes it hard to derive the accurate error bound. To simplify the problem, we neglect the error incurred by hash collisions. We note that in practice, the distribution of most attributes (queuing delay, packet interval, *etc.*) is highly skewed. Take queuing delay as an example. Only a small fraction of packets have large queuing delay, and most packets have small delay. Therefore, the first (leftmost) counter in each histogram of CellSketch has the highest chance to overflow. In the following, we assume that only the leftmost counter can overflow to derive an error upper bound. Note that without this assumption, the error bound in Theorem 5.1 also holds, but the gap between this bound and the real error will become larger. We ignore the error introduced by counter overflow, which is small and can be tuned by changing the overflow threshold.

Theorem 5.1. *Consider a histogram $\mathcal{C}_l[h]$ in CellSketch with r counters whose interval lengths follow a geometric sequence with a ratio of α . Let x be the real p -quantile ($0 < p < 1$), and suppose x falls into $\mathcal{C}_l[h][k]$. We have the relative error $\mathcal{E} := \frac{|x - \hat{x}|}{x}$ is bounded by $\mathcal{E} \leq \frac{\alpha^k - \alpha^{k-1}}{\alpha^k + \alpha^{k-1} - 2}$.*

Proof. We consider the k_{th} counter $\mathcal{C}_l[h][k]$. We suppose the interval length of $\mathcal{C}_l[h][k]$ is $\alpha^{k-1} \cdot T$. Suppose the first counter's interval starts from 0. We can calculate the left endpoint of $\mathcal{C}_l[h][k]$'s interval as $L = \sum_{i=1}^{k-1} \alpha^{i-1} \cdot T = \frac{1 - \alpha^{k-1}}{1 - \alpha} \cdot T$, and the right endpoint of its interval as $R = \sum_{i=1}^k \alpha^{i-1} \cdot T = \frac{1 - \alpha^k}{1 - \alpha} \cdot T$.

To query the p -quantile, CellSketch reports an estimated value as $\hat{x} = \frac{2}{\frac{1}{L} + \frac{1}{R}} = \frac{2LR}{L+R}$. We calculate the relative error under the following two cases.

Case 1: $x \leq \hat{x}$:

$$\mathcal{E} = \frac{\hat{x} - x}{x} = \frac{\hat{x}}{x} - 1 \leq \frac{\hat{x}}{L} - 1 = \frac{R - L}{L + R} = \frac{\alpha^k - \alpha^{k-1}}{\alpha^k + \alpha^{k-1} - 2} \quad (5)$$

Case 2: $x > \hat{x}$:

$$\mathcal{E} = \frac{x - \hat{x}}{x} = 1 - \frac{\hat{x}}{x} \leq 1 - \frac{\hat{x}}{R} = \frac{R - L}{L + R} = \frac{\alpha^k - \alpha^{k-1}}{\alpha^k + \alpha^{k-1} - 2} \quad (6)$$

We can see that in both the two cases above, we can derive the same conclusion that $\mathcal{E} \leq \frac{\alpha^k - \alpha^{k-1}}{\alpha^k + \alpha^{k-1} - 2}$ $\square$

Discussion: Theorem 5.1 gives an error upper bound of one histogram in CellSketch. We can repeatedly use this theorem in the child histograms to derive finer-grained error bounds of CellSketch. From this theorem, we can see that smaller α (ratio of the geometric series) goes with smaller relative error. However, using small α means that we must use more counters to divide the initial interval, which leads to more memory usage. In other words, there exists a trade-off between α and memory usage. In our experiments, we recommend to set $\alpha = 2$, which suffices for good performance.

2 Additional Experimental Results

We present more examples of AnoTable on estimating per-flow *packet interval* distribution in Figure 1, where we select some flows (the top- k largest flows with different k) and plot their real and estimated CDF. We conduct experiments using CAIDA dataset, and we fix the total memory of AnoTable (including CellSketch) to 100KB. We can see that for most flows (especially large flows), their estimated CDF are highly consistent with their real CDF. As AnoTable automatically allocates more memory to large flows and less memory to small flows, the estimated results for small flows might not always be very accurate. Nonetheless, we can see that even for small flows, their estimated CDF also closely match the real CDF.

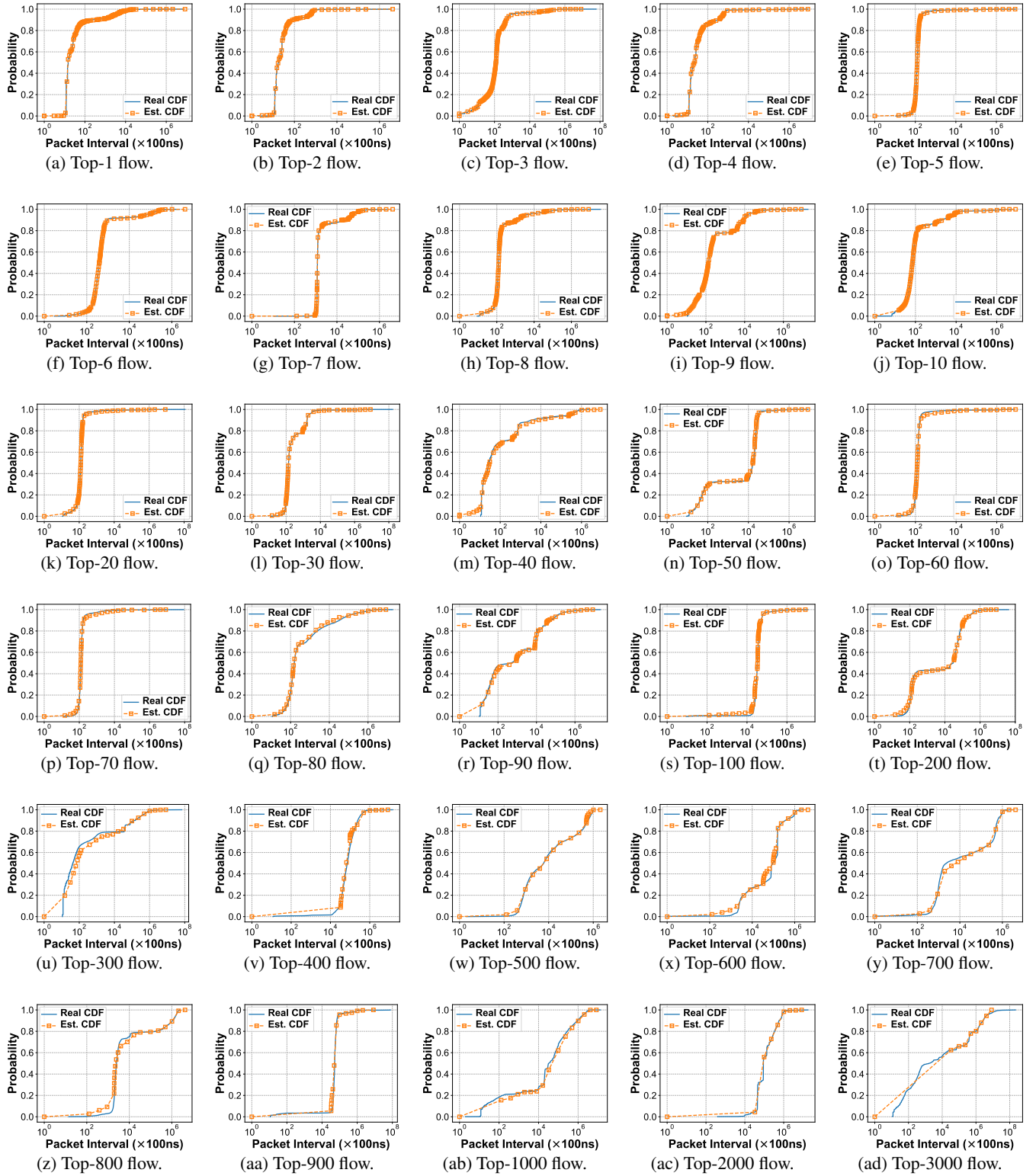


Figure 1: Real and estimated CDF (reported by AnoTable) of different flows.

References

- [1] Yinda Zhang, Zaoxing Liu, Ruixin Wang, Tong Yang, Jizhou Li, Ruijie Miao, Peng Liu, Ruwen Zhang, and Junchen Jiang. Cocosketch: high-performance sketch-based measurement over arbitrary partial key query. In *Proceedings of the 2021 ACM SIGCOMM 2021 Conference*, pages 207–222, 2021.